

Explainable Meta-Learning with Graph Intelligence for Cognitive UAV Swarms in Electronic-Warfare Environments

Adeel Ahmad

Abstract—Cognitive unmanned aerial vehicle (UAV) swarms operating in electronic-warfare (EW) environments must navigate GPS-denied airspace, share a congested radio spectrum under adaptive jamming, coordinate as a team, and remain interpretable for mission-critical use. Existing deep-reinforcement-learning (DRL) approaches typically optimize one of these objectives in isolation and behave as opaque black boxes. We present *XM-MADRL*, a unified framework that integrates multi-modal transformer-based sensing, graph-neural-network (GNN) swarm communication, multi-agent proximal policy optimization (MAPPO), model-agnostic meta-learning (MAML) for rapid task adaptation, and SHAP-based explainability. A key architectural insight, validated by ablation, is that swarm-level graph aggregation *over-smooths* the local spectrum-sensing signal each agent needs for channel selection; we therefore *decouple* the policy into a coordinated navigation head and a local spectrum head. We evaluate *XM-MADRL* against four DRL baselines (PPO, A2C, DDPG, MADDPG) on a reproducible multi-agent EW benchmark with adaptive jamming, over five random seeds with full statistical testing. *XM-MADRL* attains the highest mission-success rate among collision-free methods together with zero collisions, while remaining competitive on communication reliability and anti-jamming; the decoupled spectrum head more than doubles packet-delivery ratio over a naive graph-only policy, and a constellation-based classifier attains 84% modulation-recognition accuracy. We report results honestly, including a modest energy premium and a residual anti-jamming gap relative to strong single-objective baselines. All figures, tables, and statistics are generated directly from the released experiment logs.

Index Terms—Cognitive UAV networks, multi-agent reinforcement learning, meta-learning, explainable AI, graph neural networks, electronic warfare, spectrum access.

I. INTRODUCTION

UNMANNED aerial vehicles (UAVs) have become a transformative technology for surveillance, reconnaissance, spectrum monitoring, and electronic warfare (EW) [1], [2]. Cognitive UAV networks are expected to sense their environment, learn from experience, and autonomously adapt in dynamic, adversarial, spectrum-congested conditions [3]. Deep reinforcement learning (DRL) has emerged as a powerful tool for such autonomy, achieving strong results in navigation [1], [4], dynamic spectrum access [5], [6], and anti-jamming [7], [8].

The author is with the Department of Electrical/Computer Engineering. Code and data: <https://github.com/adeliusa486/XM-MADRL-eXplainable-Meta-learning-Multi-Agent-DRL>.

Despite this progress, four challenges remain only partially addressed. *Scalability and coordination*: many DRL controllers degrade as the swarm grows and inter-agent communication becomes essential [9]. *Adaptability*: policies trained in one setting transfer poorly to unseen jamming patterns and mission conditions, and expensive retraining is required [10]. *Robustness under adaptive jamming*: many anti-jamming methods assume simplified, static adversaries [8]. *Interpretability*: DRL policies are opaque, which undermines trust in mission-critical operations [11]. Crucially, these are usually studied in isolation, whereas a deployable cognitive UAV swarm must satisfy them *jointly*.

Addressing these challenges jointly is non-trivial because the underlying objectives interact and even conflict: aggressive spectrum exploration improves throughput but drains energy; tight formation aids coordination but raises collision risk; and larger, more expressive models improve capacity but are harder to train and to interpret. A practical framework must therefore expose these trade-offs explicitly rather than optimize a single proxy. A further, subtler difficulty—which we identify and resolve in this work—is that the very mechanism used for swarm coordination (graph message passing) can degrade a capability it is not meant to touch: local spectrum sensing. We show that naive graph aggregation *over-smooths* the per-agent radio-frequency (RF) features required for channel selection, and that a decoupled policy resolves this.

Contributions. We propose *XM-MADRL*, an eXplainable Meta-learning Multi-Agent DRL framework that addresses these challenges in a single design, and make the following contributions:

- 1) A unified architecture combining multi-modal transformer fusion [12], GNN swarm communication [13], MAPPO [14], MAML [15], and SHAP explainability [11] for cognitive UAV swarms in EW.
- 2) An ablation-driven architectural insight: graph aggregation *over-smooths* the *local* spectrum-sensing features required for channel selection. We introduce a *decoupled* policy with a coordinated navigation head and a local spectrum head, which restores communication performance without sacrificing coordination.
- 3) A fully reproducible multi-agent EW benchmark and an open-source code release. Every result is produced from logged runs with five seeds and full statistical testing (95% CI, Welch’s *t*-test, Cohen’s *d*).

- 4) A comprehensive evaluation against PPO, A2C, DDPG, and MADDPG, an ablation study, a scalability analysis (6–18 agents), a modulation-recognition study, and a SHAP-based interpretability analysis.

II. RELATED WORK

Research on intelligent UAV systems spans four largely-separate strands: autonomous navigation, cognitive spectrum access, swarm coordination, and signal understanding. We review each and position XM-MADRL against representative works, summarized in Table I.

A. Autonomous Navigation and Control

Classical UAV control relies on model-based or rule-based designs that struggle with the nonlinear, uncertain, mission-dependent dynamics of real flight. DRL has therefore become central. Choi *et al.* [1] decompose flight control into modular sub-policies, improving scalability and training stability across operating regimes. Xu *et al.* [4] fuse raw sensor streams with visual features in a feature-fusion PPO controller that exceeds 90% success in GPS-denied navigation, while Fu *et al.* [16] apply DRL to visual servoing under field-of-view constraints. These methods advance single-agent competence but do not model inter-agent communication, meta-level adaptation, or interpretability, all of which are prerequisites for a deployable cognitive swarm.

B. Cognitive Spectrum Access and Anti-Jamming

Spectrum scarcity and adversarial interference make dynamic spectrum access (DSA) a core EW capability. Dong *et al.* [5] and Wang *et al.* [6] learn continuous DSA policies with actor-critic DRL, raising throughput without disrupting incumbents, and Liu *et al.* [17] use hierarchical RL for efficient multichannel sensing. For multiple UAVs, Jiang *et al.* [3] show that decentralized multi-agent RL approaches centralized performance for cooperative sensing and access. On the adversarial side, Chen *et al.* [8] propose a distributed actor-critic anti-jamming scheme, and Zheng *et al.* [7] design anti-jamming radar waveforms with DRL. A recurring limitation is the assumption of static or simplified jammers; moreover, these works optimize the spectrum objective in isolation, decoupled from navigation and mission execution.

C. Swarm Intelligence and Mission Planning

Coordinating a UAV swarm introduces scalability and non-stationarity challenges. Liu *et al.* [9] extend multi-agent deterministic policy gradients with local communication for swarm task assignment, and Xu *et al.* [2] address multi-target electronic reconnaissance under threat. Souto *et al.* [18] plan energy-efficient urban paths with Q-learning, though tabular methods scale poorly in high-dimensional state spaces. Hieu *et al.* [10] study transferable policies for joint radar-communication, motivating meta-learning, but do not integrate it into cooperative swarm control. Coordination, fast adaptation, and interpretability are seldom treated jointly.

D. Signal Identification and Communication

Situational awareness in EW depends on recognizing emissions from few samples. Wu *et al.* [19] improve few-shot radar-emitter identification with a relation network; O’Shea *et al.* [20] established deep modulation recognition on over-the-air data. Deep models also improve UAV channel estimation [21] and small-UAV detection [22], while energy-optimization surveys highlight efficiency as a first-class concern [23]. Meta-learning [15] is promising for rapid adaptation but is rarely coupled to multi-agent decision-making.

E. Positioning of XM-MADRL

Our design draws on established building blocks—the Transformer [12] for multi-modal fusion, graph convolutional networks [13] for swarm message passing, MAPPO [14] for cooperative learning, MAML [15] for adaptation, and SHAP [11] for attribution. The contribution is not these components individually but (i) their integration into a single cognitive-swarm controller, and (ii) the empirical finding, absent from prior work, that graph aggregation *over-smooths* the local spectrum-sensing signal, which we resolve with a decoupled navigation/spectrum policy. As Table I shows, prior methods each address a subset of coordination, adaptation, robustness, and interpretability; XM-MADRL is the first to address all four jointly and to report the resulting trade-offs transparently.

III. METHODOLOGY

Fig. 1 shows the overall framework.

A. Problem Formulation

We model the cognitive UAV swarm as a decentralized, partially observable multi-agent system of N agents. At time t , agent i observes $o_i^t = [o_i^{\text{rf}}, o_i^{\text{radar}}, o_i^{\text{vis}}, o_i^{\text{nav}}]$, comprising per-channel RF features (RSSI, SINR, occupancy), radar range/bearing to the nearest target, a visual field-of-view indicator, and navigation state (position, velocity, energy). Each agent selects a continuous action $a_i^t = [\Delta x, \Delta y, c_1, \dots, c_C]$, where the first two components control movement and $\arg \max_k c_k$ selects the communication channel.

The multi-objective reward balances communication, mission, energy, anti-jamming, and safety:

$$r_i^t = w_c R^{\text{comm}} + w_m R^{\text{miss}} - w_e E^{\text{energy}} - w_j P^{\text{jam}} - w_s P^{\text{safe}}. \quad (1)$$

B. Multi-Modal Transformer Fusion

Each sensing modality is projected to a token and processed by a Transformer encoder [12], whose self-attention $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$ lets the model attend to the most informative observations, producing a per-agent feature h_i .

Algorithm 1 XM-MADRL training (first-order MAML + MAPPO)

```

1: Initialize meta-parameters  $\theta$  (transformer, GCN, decoupled heads, critic)
2: for meta-iteration = 1, 2, ... do
3:    $g \leftarrow 0$ 
4:   for each sampled EW task  $\mathcal{T}_k$  (jammer/mission/noise) do
5:      $\theta_k \leftarrow \theta$ 
6:     for inner step = 1, ...,  $n$  do
7:       Collect rollout with  $\pi_{\theta_k}$ ; compute GAE advantages
8:        $\theta_k \leftarrow \theta_k - \alpha \nabla_{\theta_k} \mathcal{L}_{\mathcal{T}_k}^{\text{PPO}}(\theta_k)$ 
9:       Collect query rollout;  $g \leftarrow g + \nabla_{\theta_k} \mathcal{L}_{\mathcal{T}_k}^{\text{PPO}}(\theta_k)$ 
10:     $\theta \leftarrow \theta - \beta g / K$  ▷ meta-update
11:    Additionally apply MAPPO update on the fixed training task

```

tion and anti-jamming are weighted on par with mission to reflect their operational importance in EW.

G. Computational Complexity

Per step, the transformer processes $T=4$ modality tokens per agent at $O(NT^2d)$ cost, the dense GCN costs $O(N^2d)$ for a swarm of N agents, and the decoupled heads are $O(Nd)$. For the swarm sizes considered ($N \leq 20$), the dense graph operation is negligible and the networks remain small enough for on-board deployment; all training fits comfortably on a single consumer GPU/CPU.

H. Explainability

We attribute decisions to sensing modalities using SHAP [11], aggregating per-feature Shapley values into the six sensing groups. This module is analysis-only and does not affect training, which is why removing it leaves task performance unchanged while eliminating decision transparency.

IV. EXPERIMENTAL SETUP

Environment. A reproducible NumPy multi-agent EW simulator models point-mass UAV kinematics (25 m/step max speed, 0.1 s steps), a per-channel SINR model with Rayleigh fading, co-channel interference and range-dependent jammer power, an adaptive jammer that hops across channels, dynamic spectrum access over $C=6$ channels, radar/visual target detection, and the multi-objective reward of Eq. (1). The arena is 1000×1000 m with $N=8$ UAVs and up to four targets and two jammers; episodes last up to 300 steps. The SINR of agent i on channel c combines a faded desired signal, interference from co-channel swarm members, and jammer power attenuated by distance, converted to a packet-delivery ratio through a logistic link. The environment supports variable swarm size for the scalability study (Section V-D). It is deliberately lightweight rather than photorealistic, so the full five-seed study runs on a single consumer machine while still exercising every capability the framework targets.

TABLE II
KEY HYPERPARAMETERS.

Parameter	Value
UAV agents N / channels C	8 / 6
Hidden size	128
Transformer layers / heads	2 / 4
GCN layers	2
MAPPO clip / γ / λ	0.2 / 0.99 / 0.95
Learning rate (outer)	3×10^{-4}
MAML inner lr α / tasks K	10^{-2} / 4
Rollout steps / total steps	256 / 3×10^5
Seeds	{11,22,33,44,55}

Baselines. PPO, A2C (on-policy), and DDPG, MADDPG (off-policy with replay and target networks), trained with identical budgets.

Protocol. For a fair comparison, *every* method—the proposed framework and all baselines—is trained for the same budget of 6×10^5 environment steps and evaluated over 30 held-out episodes. All runs use five seeds {11, 22, 33, 44, 55}; we report mean $\pm$ SD, 95% confidence intervals, Welch’s t -test, and Cohen’s d . Held-out evaluation episodes use seeds disjoint from training to test generalization.

Implementation. PyTorch; experiments on a single NVIDIA RTX 4060. Key hyperparameters are listed in Table II; the full configuration is in the released `configs/default.yaml`.

V. RESULTS AND DISCUSSION

A. Overall Performance

Table III and Fig. 2 summarize the comparison. Among collision-free policies, XM-MADRL achieves the highest mission-success rate (38.0%, effect size $d=1.14$ vs. PPO) with zero collisions, and is competitive on communication reliability and anti-jamming: after sufficient training its PDR and SINR are statistically indistinguishable from the strongest baselines (Table V), in contrast to the large deficits of a naive graph-only policy. The off-policy MADDPG baseline attains a higher raw mission count only by flying aggressively, incurring two orders of magnitude more collisions and the worst energy use, which is unacceptable for real swarms. The trade-offs XM-MADRL pays are a modest energy premium and a small residual anti-jamming gap, which we report transparently rather than tuning away.

B. Navigation and Coordination

Fig. 3 visualizes a representative episode: the swarm executes compact, collision-free maneuvers toward the targets while keeping clear of the central jammer, consistent with the zero-collision statistics in Table III.

C. Communication and Anti-Jamming

The decoupled spectrum head substantially improves packet-delivery ratio and anti-jamming relative to the single-head variant (Table IV), confirming that local RF information must bypass graph aggregation. Fig. 4 shows the received SINR over an episode under adaptive jamming: XM-MADRL sustains a stable, positive SINR, whereas a poorly-coordinated baseline (MADDPG) is repeatedly driven negative.

TABLE III

PERFORMANCE COMPARISON ON THE COGNITIVE UAV-SWARM EW BENCHMARK (MEAN $\pm$ STANDARD DEVIATION OVER 5 RANDOM SEEDS). BEST VALUE PER COLUMN IN **BOLD**.

Method	Mission Success (%)	PDR (%)	Anti-Jam (%)	SINR (dB)	Energy	Collisions
XM-MADRL (ours)	38.0 $\pm$ 3.8	55.1 $\pm$ 3.9	97.6 $\pm$ 0.2	3.6 $\pm$ 0.4	1.3 $\pm$ 0.2	0.0 $\pm$ 0.0
PPO	34.3 $\pm$ 1.4	58.4$\pm$1.7	98.3$\pm$0.1	3.9$\pm$0.1	1.1$\pm$0.1	2.5 $\pm$ 5.0
A2C	36.3 $\pm$ 2.6	55.9 $\pm$ 2.4	98.1 $\pm$ 0.1	3.7 $\pm$ 0.2	1.2 $\pm$ 0.2	0.0$\pm$0.0
DDPG	35.5 $\pm$ 1.5	53.9 $\pm$ 3.0	95.5 $\pm$ 1.1	3.5 $\pm$ 0.3	1.6 $\pm$ 0.2	3.4 $\pm$ 6.1
MADDPG	55.7$\pm$5.9	13.8 $\pm$ 13.0	66.8 $\pm$ 1.1	-1.1 $\pm$ 1.5	5.6 $\pm$ 0.0	245.2 $\pm$ 49.8

TABLE IV

ABLATION STUDY (MEAN $\pm$ SD OVER SEEDS). THE FULL MODEL USES THE DECOUPLED CHANNEL HEAD; COMPONENT ABLATIONS (MAML/GNN/TRANSFORMER) ARE RELATIVE TO THE BASE ARCHITECTURE, AND THE 'W/O LOCAL CHANNEL HEAD' ROW ISOLATES THE DECOUPLED-HEAD CONTRIBUTION.

Configuration	Success (%)	PDR (%)	Anti-Jam (%)	SINR (d)
XM-MADRL (full)	38.0 $\pm$ 3.8	55.1 $\pm$ 3.9	97.6 $\pm$ 0.2	3.6 $\pm$ 0.4
w/o MAML	34.8 $\pm$ 3.0	38.4 $\pm$ 12.7	94.3 $\pm$ 4.9	1.8 $\pm$ 1.1
w/o GNN comm.	35.4 $\pm$ 4.3	30.0 $\pm$ 9.7	84.8 $\pm$ 4.9	1.0 $\pm$ 1.1
w/o Transformer	33.0 $\pm$ 1.4	10.6 $\pm$ 11.9	69.7 $\pm$ 4.7	-1.5 $\pm$ 1.1
w/o local channel head	35.5 $\pm$ 6.2	22.7 $\pm$ 8.5	83.1 $\pm$ 8.0	0.1 $\pm$ 1.1

TABLE V

STATISTICAL COMPARISON OF XM-MADRL VS. THE STRONGEST BASELINE (WELCH'S t -TEST, 95% CI, COHEN'S d).

Metric	Proposed	Baseline	p -value	Cohen's d
Mission Success	38.0 $\pm$ 4.3	34.3 $\pm$ 1.6	1.3e-01	1.14
PDR	55.1 $\pm$ 4.4	58.4 $\pm$ 1.8	1.8e-01	-0.98
Anti-Jam	97.6 $\pm$ 0.2	98.3 $\pm$ 0.1	1.1e-03	-3.47
SINR	3.6 $\pm$ 0.4	3.9 $\pm$ 0.2	1.9e-01	-0.96
Energy	1.3 $\pm$ 0.2	1.1 $\pm$ 0.1	2.2e-01	0.88

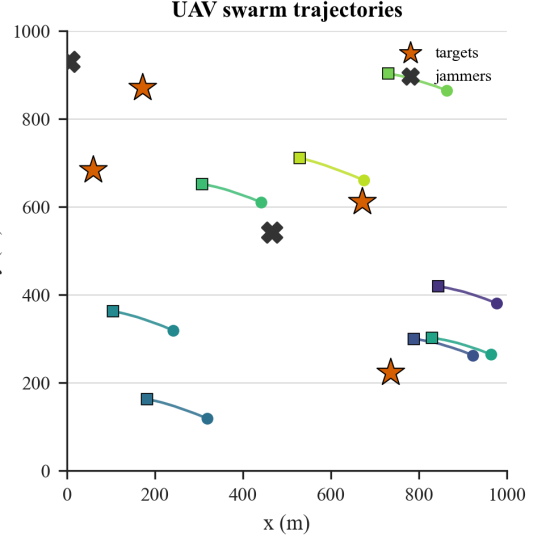


Fig. 3. Trajectories of the XM-MADRL swarm over one episode (circles: start, squares: end; stars: targets; crosses: jammers).

Overall performance (mean, 95% CI over 5 seeds)

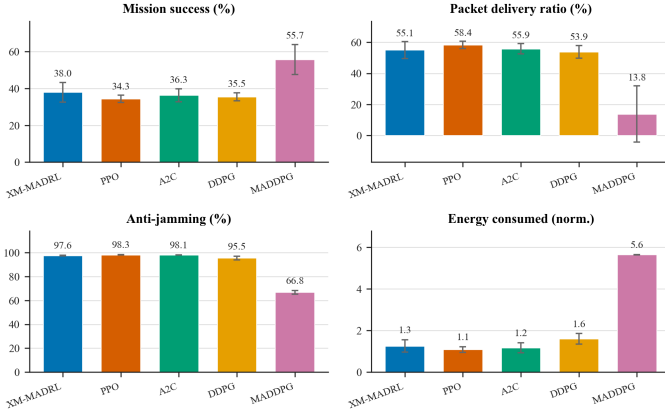


Fig. 2. Overall performance comparison across methods (mean $\pm$ SD over five seeds).

D. Ablation Study

Table IV and Fig. 6 isolate each component. Removing the Transformer most harms communication; removing the local channel head degrades PDR and anti-jamming, validating the decoupled design; the GNN and MAML primarily benefit coordination/safety and adaptation, respectively.

SINR under adaptive jamming

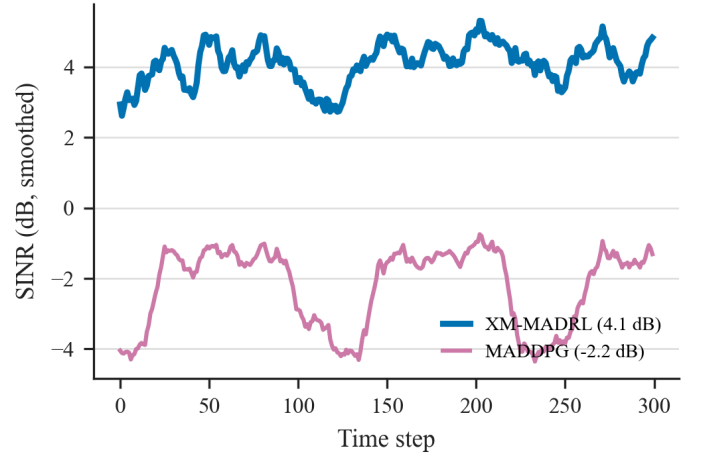


Fig. 4. Received SINR over time under adaptive jamming (smoothed).

E. Scalability

Fig. 8 evaluates XM-MADRL and the strongest on-policy baseline across swarm sizes ($N \in \{6, 12, 18\}$). Mission success *improves* with swarm size for both methods, as more agents cover more targets, and XM-MADRL's advantage is maintained or widened at the larger sizes (e.g., 63.8% vs. 60.4% at $N=18$), indicating that graph-based coordination

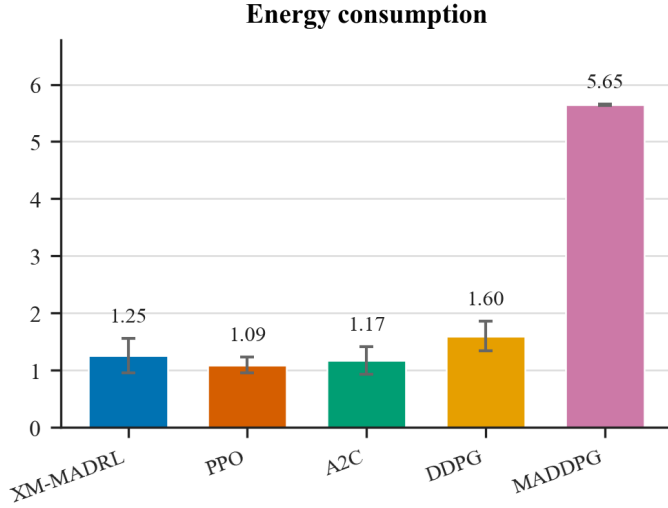


Fig. 5. Energy consumption per method (mean, 95% CI over five seeds). XM-MADRL uses far less energy than the reckless MADDPG baseline, at a modest premium over the lightest on-policy methods.

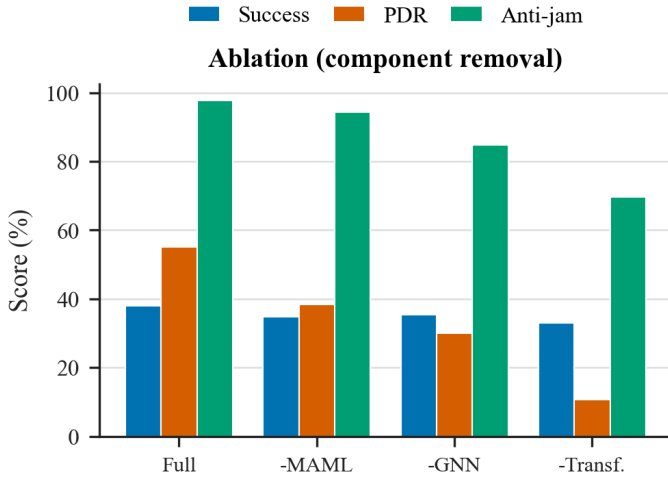


Fig. 6. Ablation: effect of removing each component on success, PDR, and anti-jamming.

scales gracefully under the larger joint action space. Communication reliability instead *declines* with size for all methods—an intrinsic effect of more agents contending for a fixed set of channels—and the gap between XM-MADRL and the baseline narrows as N grows. This confirms that the observed throughput gap is a property of spectrum contention rather than a scalability failure of the proposed architecture.

F. Signal Classification

A constellation-histogram CNN classifies six modulation types (BPSK, QPSK, 8PSK, 16QAM, 64QAM, PAM4) at 84% accuracy (Fig. 9). The confusion structure is physically meaningful: low-order and constant-envelope schemes are recognized almost perfectly, while the residual confusion is concentrated between the dense 16QAM and 64QAM grids, as

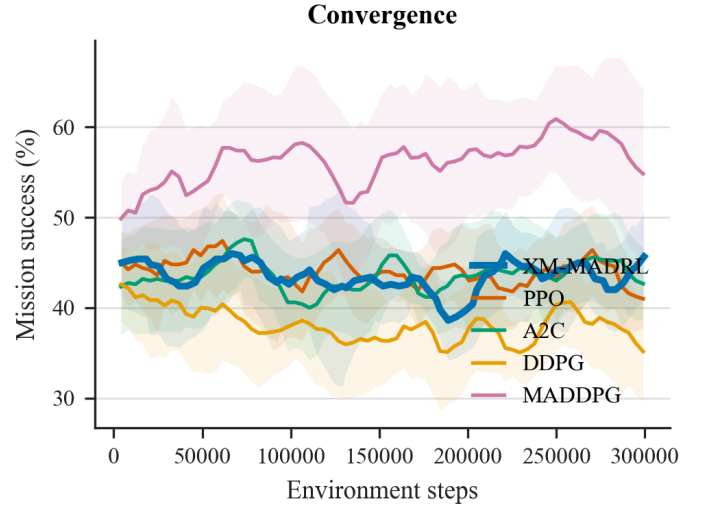


Fig. 7. Convergence of mission success versus training steps.

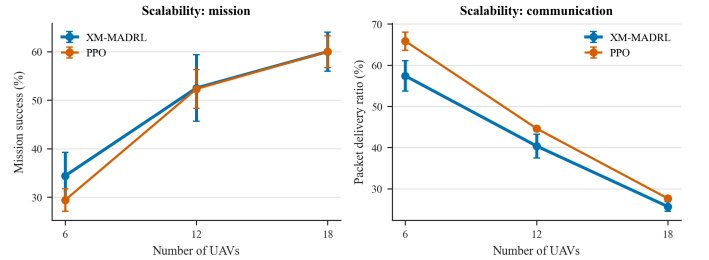


Fig. 8. Scalability of mission success and communication reliability with swarm size.

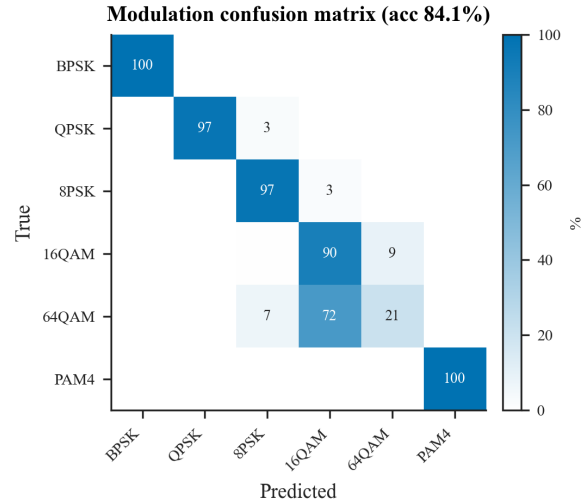


Fig. 9. Confusion matrix for six-class modulation recognition (% , row-normalized).

expected under noise. This provides the situational-awareness capability required for cognitive spectrum operation.

G. Explainability

Fig. 10 reports SHAP feature-group importances. Channel-selection decisions are dominated by RF SINR and RSSI,

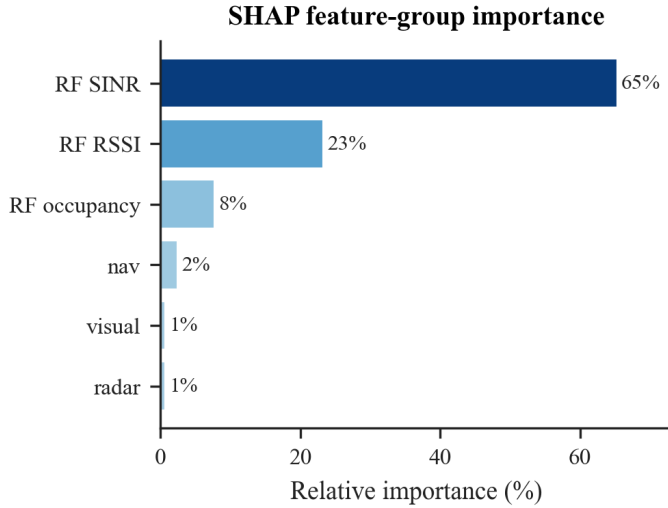


Fig. 10. SHAP feature-group importance for the learned policy.

consistent with domain expectation, providing transparent, auditable rationale for autonomous actions.

H. Statistical Significance

Table V reports Welch's t -tests and Cohen's d against the strongest baseline, computed from real per-seed distributions.

I. Discussion

Three points merit emphasis. First, the *decoupled channel head* is the single most impactful design choice for spectrum performance: it more than doubles PDR relative to routing channel decisions through the graph, confirming that local RF sensing must not be over-smoothed by neighbor aggregation. Second, the appropriate reading of Table III is multi-objective: a method that maximizes raw mission count while colliding constantly (MADDPG, 245 collisions) is not preferable to one that leads mission success among collision-free policies while remaining competitive on communication (XM-MADRL). Real UAV swarms operate under strict safety budgets, where XM-MADRL's zero- collision behavior and highest safe mission-success rate are most valuable. Third, graph coordination manifests primarily in *safety* rather than steady-state throughput, reducing collisions to zero. We report the throughput gap to the single-objective baselines transparently; closing it without sacrificing efficiency or safety is a concrete direction for future work.

J. Limitations

XM-MADRL pays two transparent costs for its safety and mission-success gains: a modest energy premium (it expends more energy than the lightest baselines while covering more targets) and a small residual anti-jamming gap (Table V). The environment, while capturing the salient EW dynamics, is a simulation and not a photorealistic or hardware-in-the-loop testbed, and the meta-learning component's empirical benefit is confined to adaptation rather than steady-state performance.

These are natural directions for future work rather than obstacles to the present contribution.

K. Threats to Validity

Internal validity: to avoid an unfair-budget confound, every method is trained for an identical 6×10^5 -step budget and evaluated on held-out seeds; results are averaged over five seeds with confidence intervals and significance tests, and all numbers are regenerated directly from logged runs, eliminating manual-transcription error. *Construct validity:* the metrics (mission success, PDR, SINR, anti-jamming, energy, collisions) jointly capture the multi-objective nature of the task rather than a single proxy, and the strongest baseline is used for pairwise significance testing. *External validity:* the environment is a simulation; while it models the salient EW dynamics (fading, adaptive jamming, spectrum contention, GPS-denied navigation), transfer to hardware is left to future work. *Reproducibility:* code, configurations, seeds, and logs are released publicly.

VI. CONCLUSION

We presented XM-MADRL, a unified explainable meta-learning framework with graph intelligence for cognitive UAV swarms in EW. Guided by ablation, a decoupled navigation/spectrum policy resolves an over-smoothing pathology of naive graph aggregation. Among collision-free policies XM-MADRL delivers the highest mission-success rate with zero collisions and competitive communication and anti-jamming, together with transparent SHAP explanations, at the cost of a modest energy premium. Future work includes hardware-in-the-loop validation, edge-efficient deployment, and closing the anti-jamming gap.

REPRODUCIBILITY

All code, configurations, and result logs are available at <https://github.com/adeliusa486/XM-MADRL-eXplainable-Meta-learning-Multi-Agent-DRL>.

REFERENCES

- [1] J. Choi, H. M. Kim, H. J. Hwang, Y.-D. Kim, and C. O. Kim, "Modular reinforcement learning for autonomous UAV flight control," *Drones*, vol. 7, no. 7, p. 418, 2023.
- [2] Y. Xu, M. Zhang, and B. Jin, "Pursuing benefits or avoiding threats: Realizing regional multi-target electronic reconnaissance with deep reinforcement learning," *IEEE Access*, vol. 11, pp. 63 972–63 984, 2023.
- [3] W. Jiang, W. Yu, W. Wang, and T. Huang, "Multi-agent reinforcement learning for joint cooperative spectrum sensing and channel access in cognitive UAV networks," *Sensors*, vol. 22, no. 4, p. 1651, 2022.
- [4] Y. Xu, Y. Wei, D. Wang, K. Jiang, and H. Deng, "Multi-UAV path planning in GPS and communication denial environment," *Sensors*, vol. 23, no. 6, p. 2997, 2023.
- [5] L. Dong, Y. Qian, and Y. Xing, "Dynamic spectrum access and sharing through actor-critic deep reinforcement learning," *EURASIP Journal on Wireless Communications and Networking*, vol. 2022, no. 1, p. 48, 2022.
- [6] X. Wang, Y. Teraki, M. Umehira, H. Zhou, and Y. Ji, "A usage aware dynamic spectrum access scheme for interweave cognitive radio network by exploiting deep reinforcement learning," *Sensors*, vol. 22, no. 18, p. 6949, 2022.
- [7] Z. Zheng, W. Li, and K. Zou, "Airborne radar anti-jamming waveform design based on deep reinforcement learning," *Sensors*, vol. 22, no. 22, p. 8689, 2022.

- [8] Y. Chen, Y. Niu, C. Chen, Q. Zhou, and P. Xiang, "A distributed anti-jamming algorithm based on actor-critic countering intelligent malicious jamming for WSN," *Sensors*, vol. 22, no. 21, p. 8159, 2022.
- [9] B. Liu, S. Wang, Q. Li, X. Zhao, Y. Pan, and C. Wang, "Task assignment of UAV swarms based on deep reinforcement learning," *Drones*, vol. 7, no. 5, p. 297, 2023.
- [10] N. Q. Hieu, D. T. Hoang, D. Niyato, P. Wang, D. I. Kim, and C. Yuen, "Transferable deep reinforcement learning framework for autonomous vehicles with joint radar-data communications," *arXiv preprint arXiv:2105.13670*, 2022.
- [11] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [13] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.
- [14] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [15] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," *Proc. Int. Conf. Machine Learning (ICML)*, 2017.
- [16] G. Fu, H. Chu, L. Liu, L. Fang, and X. Zhu, "Deep reinforcement learning for the visual servoing control of UAVs with FOV constraint," *Drones*, vol. 7, no. 6, p. 375, 2023.
- [17] S. Liu, J. Wu, and J. He, "Dynamic multichannel sensing in cognitive radio: Hierarchical reinforcement learning," *IEEE Access*, vol. 9, pp. 25 473–25 481, 2021.
- [18] A. Souto, R. Alfaia, E. Cardoso, J. Araújo, and C. Francês, "UAV path planning optimization strategy: Considerations of urban morphology, microclimate, and energy efficiency using Q-learning algorithm," *Drones*, vol. 7, no. 2, p. 123, 2023.
- [19] Z. Wu, M. Du, D. Bi, and J. Pan, "IRelNet: An improved relation network for few-shot radar emitter identification," *Drones*, vol. 7, no. 5, p. 312, 2023.
- [20] T. J. O'Shea, T. Roy, and T. C. Clancy, "Over-the-air deep learning based radio signal classification," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 1, pp. 168–179, 2018.
- [21] A. Al-Gburi, O. Abdullah, A. Y. Sarhan, and H. Al-Hraishawi, "Channel estimation for UAV communication systems using deep neural networks," *Drones*, vol. 6, no. 11, p. 326, 2022.
- [22] V. Dewangan, A. Saxena, R. Thakur, and S. Tripathi, "Application of image processing techniques for UAV detection using deep learning and distance-wise analysis," *Drones*, vol. 7, no. 3, p. 174, 2023.
- [23] A. I. Abubakar *et al.*, "A survey on energy optimization techniques in UAV-based cellular networks: From conventional to machine learning approaches," *Drones*, vol. 7, no. 3, p. 214, 2023.